What is the difference between "Gaussian elimination" (GE) vs "Gauss-Jordan elimination" (GJE); when do we use GE vs GJE?

Answer:

The primary difference between Gaussian elimination and Gauss-Jordan elimination lies in their end goal and, consequently, the number of steps they require.

The Core Difference: The Target Matrix Form

The fundamental distinction is the form of the matrix you are trying to achieve.

Gaussian Elimination (GE) aims to transform a matrix into **Row Echelon Form (REF)**. A matrix is in row echelon form when it satisfies a few conditions: all rows consisting entirely of zeros are at the bottom, and for the non-zero rows, the first non-zero number from the left (called the leading entry or pivot) is to the right of the pivot of the row above it. This process effectively creates a “staircase of zeros” structure at the bottom left of the matrix, where all entries below the pivots (leading entries) are zero.

Gauss-Jordan Elimination (GJE) goes a step further. It transforms a matrix into **Reduced Row Echelon Form (RREF)**. A matrix in RREF meets all the conditions for REF, but with two additional, stricter rules. First, every leading entry (pivot) must be 1. Second, each leading 1 must be the *only* non-zero entry in its entire column. This means you must create zeros not only *below* each pivot, but also *above* them.

In short, Gaussian elimination stops once the matrix is in row echelon form, while Gauss-Jordan elimination continues the process until it reaches the more specific reduced row echelon form.

When to Use Each Method:

The choice between GE and GJE depends on the specific task you are trying to accomplish.

You would typically use **Gaussian Elimination (GE)** for a few main reasons. It is computationally faster, requiring fewer row operations. This makes it the preferred method for solving a system of linear equations by hand or for numerical algorithms where efficiency is critical. It is also the method to use when you need to find the determinant of a square matrix, as you only need to get it into a triangular form (like REF) to multiply the diagonal entries.

On the other hand, you would use **Gauss-Jordan Elimination (GJE)** for other specific applications. Its most classic use is for finding the inverse of a non-singular matrix. To do this, you augment a square matrix $\boldsymbol{A}$ with the identity matrix $\boldsymbol{I}$ to form $[\boldsymbol{A} \mid \boldsymbol{I}]$ and then perform GJE until the left side becomes the identity matrix. The right side will then be the inverse, $[\boldsymbol{I} \mid \boldsymbol{A}^{-1}]$. While it takes more steps, some people prefer it for solving systems of equations because the final answer is immediately obvious and doesn't require the extra step of back substitution, which can be a source of arithmetic errors.

To summarize, Gaussian elimination is the quicker, more direct path to solving a system of equations, while Gauss-Jordan elimination is a more complete process that yields a unique, simpler final matrix form (RREF) that is essential for tasks like finding a matrix inverse.

Scenario 1

Let's see how this works with a simple system of linear equations: $\begin{cases} x + 3y = 7 \\ 2x + 4y = 10 \end{cases}$

We start by writing this as an augmented matrix: $\begin{bmatrix} 1 & 3 & 7 \\ 2 & 4 & 10 \end{bmatrix}$

Our goal is to use elementary row operations to simplify this.

1. Applying Gaussian Elimination (to get REF)

The only step needed is to create a zero in the bottom-left position (below the first pivot). We can do this by replacing R_2 (Row 2) with $R_2 - 2R_1$ (Row 2 - 2 * Row 1) to get $\begin{bmatrix} 1 & 3 & 7 \\ 0 & -2 & -4 \end{bmatrix}$

At this point, Gaussian elimination is complete. The matrix is in Row Echelon Form (REF). To find the solution, we use a process called back substitution. The second row tells us that $-2y = -4$, which means $y = 2$. We then substitute this value back into the first row's equation, $x + 3y = 7$, to get $x + 3(2) = 7$, which simplifies to $x + 6 = 7$, so $x = 1$.

2. Applying Gauss-Jordan Elimination (to get RREF)

We start from the REF we achieved above and continue the process: $\begin{bmatrix} 1 & 3 & 7 \\ 0 & -2 & -4 \end{bmatrix}$

First, we make the pivots into 1's. We can scale R_2 (Row 2) by $-1/2$: $\begin{bmatrix} 1 & 3 & 7 \\ 0 & 1 & 2 \end{bmatrix}$

Now, we must create a zero *above* the pivot in the second column. We do this by replacing R_1 (Row 1) with $R_1 - 3R_2$ (Row 1 - 3 * Row 2): $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$

Now, the process is complete. The matrix is in Reduced Row Echelon Form (RREF). The advantage here is that no back substitution is needed. The solution can be read directly from the last column of the matrix. The first row tells us $1x + 0y = 1$, so $x = 1$. The second row tells us $0x + 1y = 2$, so $y = 2$.

Scenario 2

1. When you use Gaussian elimination, you stop once you reach Row Echelon Form (REF). For a system with a unique solution, the augmented matrix will look something like this: $\begin{bmatrix} a & b & c & p \\ 0 & d & e & q \\ 0 & 0 & f & r \end{bmatrix}$

This corresponds to the system of equations: $\begin{cases} ax + by + cz = p \\ \quad dy + ez = q \\ \qquad fz = r \end{cases}$

As you can see, you have the value for z directly (z = r/f). However, to find y and x, you need to perform back substitution. You take the value of z, plug it into the second equation to solve for y, and then plug the values of both y and z into the first equation to solve for x. This is an extra computational step that can introduce arithmetic errors.

2. When you use Gauss-Jordan elimination, you continue the process until you reach Reduced Row Echelon Form (RREF). For the same system, the final augmented matrix will look like this: $\begin{bmatrix} 1 & 0 & 0 & p \\ 0 & 1 & 0 & q \\ 0 & 0 & 1 & r \end{bmatrix}$

This corresponds to the system of equations: $\begin{cases} x & = p \\ y & = q \\ z & = r \end{cases}$ Here, the simplification is clear. The solution is presented explicitly. You can simply read the values for each variable directly from the last column of the augmented matrix without any further calculation.

Scenario 3

Consider the following system with variables x_1, x_2, x_3, and x_4:

$$\begin{cases} x_1 + 2x_2 + 3x_3 + 4x_4 & = 10 \\ 2x_1 + 4x_2 + 7x_3 + 9x_4 & = 23 \\ 3x_1 + 6x_2 + 10x_3 + 13x_4 & = 33 \end{cases}$$

1. Solving with Gaussian Elimination (GE)

Step 1: Write the Augmented Matrix $\begin{bmatrix} 1 & 2 & 3 & 4 & 10 \\ 2 & 4 & 7 & 9 & 23 \\ 3 & 6 & 10 & 13 & 33 \end{bmatrix}$

Step 2: Reduce to Row Echelon Form (REF) We need to create zeros below the first pivot (the 1 in the top-left in column 1).

- Perform $R_2 \rightarrow R_2 - 2*R_1$: $\begin{bmatrix} 1 & 2 & 3 & 4 & 10 \\ 0 & 0 & 1 & 1 & 3 \\ 3 & 6 & 10 & 13 & 33 \end{bmatrix}$
- Perform $R_3 \rightarrow R_3 - 3*R_1$: $\begin{bmatrix} 1 & 2 & 3 & 4 & 10 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 & 3 \end{bmatrix}$

  Now, we create a zero below the second pivot (the 1 in the second row, column 3).
- Perform $R_3 \rightarrow R_3 - R_2$: $\begin{bmatrix} 1 & 2 & 3 & 4 & 10 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

This matrix is now in Row Echelon Form (REF). The GE process is complete.

Step 3: Identify Parameters and Use Back Substitution The columns for x_1 and x_3 are pivot columns, while the columns for x_2 and x_4 are non-pivot columns. So x_2 and x_4 are assigned as parameters.

Let $x_2 = s$ and $x_4 = t$

Now, convert the REF matrix back to equations and solve using back substitution, starting from the bottom non-zero row.

- From Row 2: $x_3 + x_4 = 3$
  - Substitute $x_4 = t$: $x_3 + t = 3 \Rightarrow x_3 = 3 - t$
- From Row 1: $x_1 + 2x_2 + 3x_3 + 4x_4 = 10$

- Substitute the expressions for x_2, x_3, and x_4:
  $x_1 + 2(s) + 3(3 - t) + 4(t) = 10$
  $x_1 + 2s + 9 - 3t + 4t = 10$
  $x_1 + 2s + t = 1$
  $x_1 = 1 - 2s - t$

Step 4: State the General Solution (from GE) The general solution is:

- $x_1 = 1 - 2s - t$
- $x_2 = s$
- $x_3 = 3 - t$
- $x_4 = t$ (where s and t are any real numbers).

2. Solving with Gauss-Jordan Elimination (GJE)

The goal of GJE is to transform the matrix into Reduced Row Echelon Form (RREF), which allows for a direct solution.

Step 1: Start from REF We begin with the REF matrix we found using GE in the first part: $\begin{bmatrix} 1 & 2 & 3 & 4 & 10 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Step 2: Reduce to Reduced Row Echelon Form (RREF) We must now create zeros *above* the pivots (leading entries). The second pivot is in the second row. We need to eliminate the 3 above it.

- Perform $R_1 \to R_1 - 3*R_2$: $\begin{bmatrix} 1 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

This matrix is now in Reduced Row Echelon Form (RREF). All pivots are 1, and they are the only non-zero entries in their respective columns. The GJE process is complete. Since there are non-pivot columns in the RREF, we need to introduce parameters and find the general solution. In this case, we cannot read the solutions of the system directly from the last column of the RREF.

Step 3: Interpret the RREF Directly The pivot and non-pivot columns in the RREF is consistent with those of any REF of the augmented matrix. So again x_2 and x_4 are assigned as parameters.

- Let $x_2 = s$ and $x_4 = t$

Now, convert the RREF matrix to equations. The simplified form allows us to solve for x_1 and x_3 directly in terms of the parameters s and t, without substitution between equations.

- From Row 1: $x_1 + 2x_2 + x_4 = 1$
  - Substitute parameters: $x_1 + 2s + t = 1 \Rightarrow x_1 = 1 - 2s - t$
- From Row 2: $x_3 + x_4 = 3$
  - Substitute parameters: $x_3 + t = 3 \Rightarrow x_3 = 3 - t$

Step 4: State the General Solution (from GJE) The general solution is:

- $x_1 = 1 - 2s - t$

- $x_2 = s$
- $x_3 = 3 - t$
- $x_4 = t$

Both methods produce the exact same general solution. The key difference is that GJE requires more row operations on the matrix but simplifies the final step, eliminating the need for back substitution.